AVIATION · DATA ENGINEERING · PRE-REGISTERED STUDY

When IndiGo broke

What happened to Indian air travel when the country's biggest airline lost control of its crew roster in December 2025.

October 2026 · Sanjana Maini · sanjanamaini.github.io/indigo

THE FINDING

> *When IndiGo's operations collapsed in December 2025, about two million passengers did not fly at all. Other airlines did not take them: they had not added their usual December seats and their planes were already full. Routes that only IndiGo flew lost twice as much traffic as routes it shared, never caught up through July 2026, and IndiGo itself ended the period with a record share of the market.*

2.11m
passengers who did not fly in December 2025
`q1_market_lost`

−12%
share of IndiGo's shortfall carried by rivals
`q1_absorption_share`

−15.6 pts
December change on routes only IndiGo flew
`h_mean_indigo_only`

67.4%
IndiGo's share in July 2026, a record
`q4_indigo_share_max`

1 What happened

On 1 November 2025 the second phase of India's revised flight duty rules for pilots took effect: a night window from midnight to 6 a.m., at most two night landings where six had been allowed, and 48 hours of weekly rest instead of 36. IndiGo, which flew 63.6% of domestic passengers in November, had not planned enough crew. Its on-time performance, 65.0% on average from 17 to 30 November, fell to 20.7% on 3 December.

source: daily.csv; numbers.json: d_indigo_otp_*

DATE	EVENT
1 Nov 2025	Phase two of the duty rules takes effect.
2 to 9 Dec	Mass cancellations; the government counts 2,507 cancelled and 1,852 delayed IndiGo flights on 3 to 5 December alone.
5 to 6 Dec	The regulator grants IndiGo temporary relief from the night-duty rules, to 10 February 2026, and issues show-cause notices to its chief executive and accountable manager; the government caps fares.
9 to 10 Dec	IndiGo's winter schedule is cut by 10%, with the slots offered to other airlines.
Jan 2026	IndiGo surrenders about 700 airport slots for January to March; a penalty of about ₹22 crore follows.
10 Feb 2026	The temporary relief ends; IndiGo says it has enough pilots.

Sources: Press Information Bureau (DGCA findings, 2026); Wikipedia, "2025 IndiGo scheduling crisis"; Business Standard and Deccan Herald reporting, December 2025 to February 2026.

2 Data, and the problem at its centre

DATASET **Four sources, joined in a DuckDB warehouse**

ROUTES	DGCA monthly passengers for every city pair, all airlines together, since 2015 (via Vonter/india-aviation-traffic, ODbL)
AIRLINES	DGCA monthly passengers, departures and seat-km for every airline, all routes together
DAYS	Ministry of Civil Aviation daily summaries: departures, passengers, on-time performance by airline (4 to 6 December not archived)
NETWORK	Which airlines flew which route, reconstructed from Wikipedia airport articles as they stood at 23:59 UTC on 30 November 2025 (and 2024, for the placebo)

IN PLAIN WORDS

The study needs to know, for every route, whether a second airline flew it. India publishes passengers *per route* and passengers *per airline*, but never both together. So the route network had to be rebuilt. Every Indian airport's Wikipedia article lists, for each airline, the destinations it serves, with "begins" and "ends" dates, and every past version of the article is kept. Reading the 684 routes from the versions in force on 30 November gives the network the day before the meltdown.

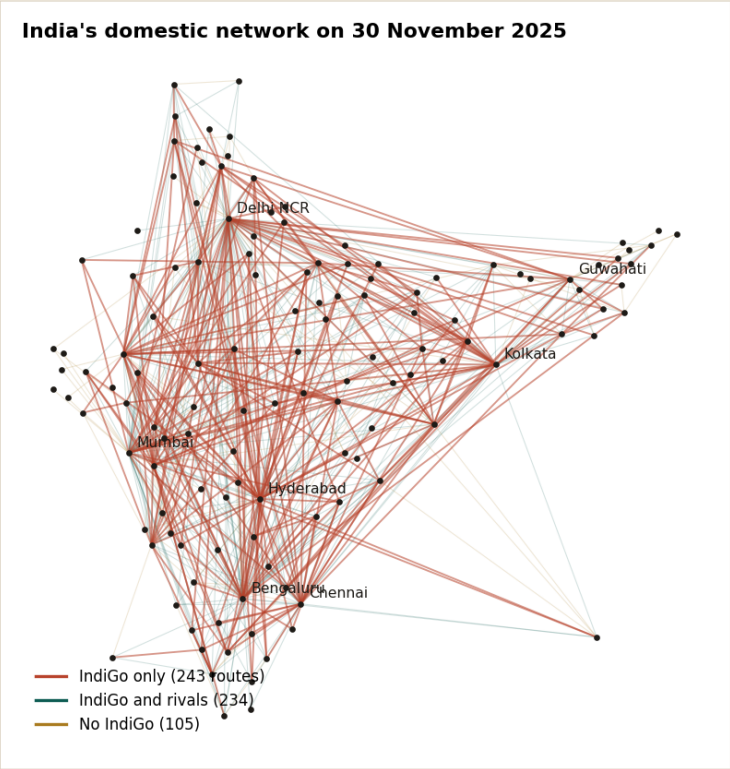


FIGURE 0.1 The domestic network on 30 November 2025: routes where IndiGo was the only airline listed (red), shared with rivals (green) and without IndiGo (ochre).

How good is the rebuilt network? It contains all 456 DGCA routes with 3,000 or more passengers that month, covering 99.9% of passengers, and 85.1% of its routes carried real traffic. What it can get wrong is *who* flies a route: if an airline is missing from both airports' articles, a shared route looks like an IndiGo-only one. A spot check of the three largest IndiGo-only routes against current airline pages found rival service on two (Hyderabad-Kolkata and Ahmedabad-Hyderabad). That error makes the two groups look more alike, so the gap reported below is, if anything, too small; and no single route in this report is claimed to be a monopoly.

DGCA RENAMED INDIA'S CITIES IN JANUARY 2026

From January 2026, "MUMBAI" became "MUMBAI (MUMBAI)", "TRIVANDRUM" became "THIRUVANANTHAPURAM" and "VIJAYAWADA" became the misspelt "VIJAYWADA"; others gained trailing spaces. Left alone, half the network would appear to vanish the month after the crisis. A mapping table repairs every label, and a test checks that no busy city changes by more than 50% across the switch. Delhi NCR, Mumbai and Goa are treated as one market each, so a passenger moving to the new Navi Mumbai airport (opened 25 December 2025) is not counted as lost.

Checks. The aggregated files match DGCA's own workbooks row for row in December 2024, November 2025 and December 2025. DGCA's two publications disagree by 0.6% in January and February 2025; that is recorded and tested

rather than hidden. The warehouse runs 18 data tests on every build.

3 Where did the passengers go?

> IN PLAIN WORDS
>
> Expected December traffic for each airline is its November 2025 traffic multiplied by its usual November-to-December rise, averaged over 2022, 2023 and 2024. The shortfall is expected minus actual.

IndiGo was expected to carry 10.40 million passengers in December and carried 8.52 million: a shortfall of 1.88 million (1.75 to 1.97 million depending on the base year). Every other airline together was expected to carry 6.02 million and carried 5.78 million, about 0.23 million *fewer*. Rivals absorbed −12% of IndiGo's shortfall (range −26% to +5%): nothing measurable. About 2.11 million passengers (1.66 to 2.42 million) did not fly.

source: tables/national_*.csv

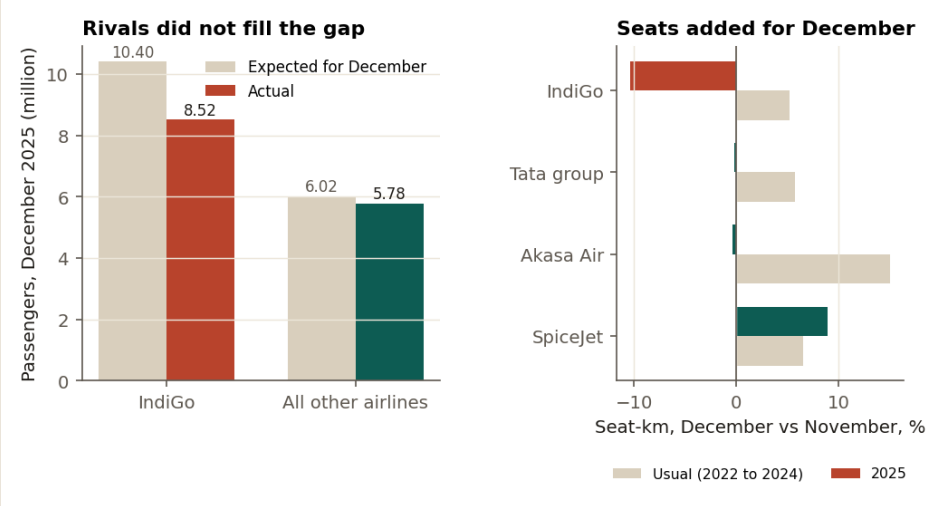


FIGURE 0.2 Left: December 2025 passengers against the seasonal expectation. Right: seat-kilometres added from November to December, 2025 against the 2022 to 2024 average.

Why rivals could not help. In a normal December the Tata group (Air India and Air India Express) adds about +5.7% seat-km over November; in 2025 it added −0.2%. Akasa Air usually adds about 15%; in 2025 its seats fell slightly, and its load factor reached 95.3% against 90.3% in past Decembers. The airlines that could have taken IndiGo's passengers had no spare seats to sell. IndiGo's own seat-km fell −10.4% where it usually rises +5.2%.

source: tables/load_factors_december.csv

> ANALOGY · IN REAL LIFE
>
> A city's biggest bakery shuts for a week before a festival. The other bakeries are already baking at full capacity for the festival rush, so they cannot bake more. Most of the bakery's customers do not buy bread elsewhere; they go without.

4 The daily shape

The lowest archived day is 7 December 2025: 2,705 departures (81.5% of the late-November average) and 382,225 departing passengers (72.9%). The worst days, 4 to 6 December, are missing from the archive, so the true trough was deeper. Departures were back above 95% of normal by 17 December 2025.

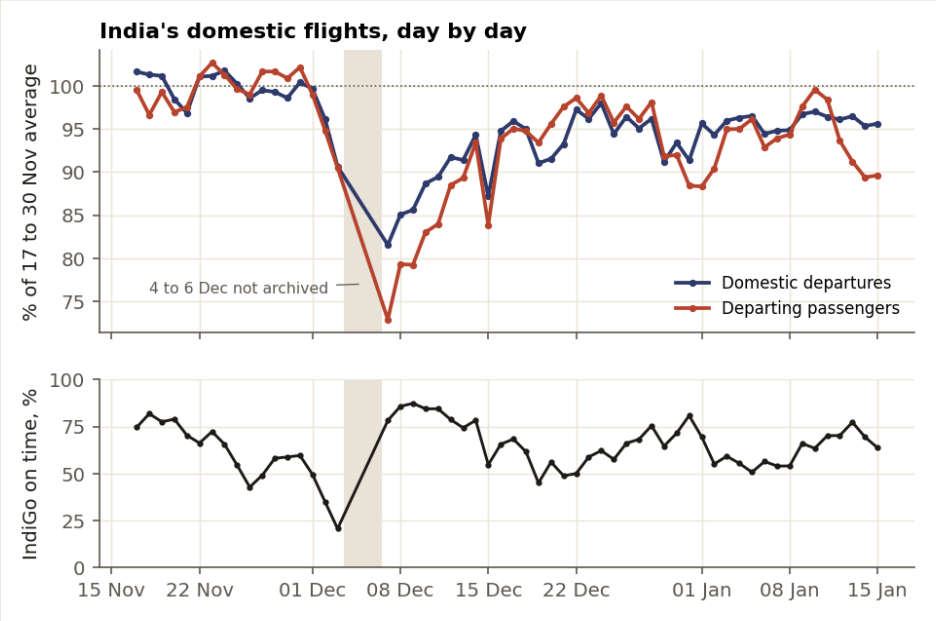


FIGURE 0.3 Domestic departures and passengers against the 17 to 30 November average, and IndiGo's on-time performance. The shaded days are not in the archive.

5 Did it matter whether a route had a second airline?

> IN PLAIN WORDS
>
> Each route is compared with itself. Its year-on-year growth in December 2025 is set against its own average year-on-year growth in October and November, which removes seasonality and each route's trend. Routes are grouped by who flew them on 30 November 2025, and group averages are weighted by passengers. The same calculation for December 2024, with the November 2024 network, is the placebo: nothing happened then, so it should show nothing.

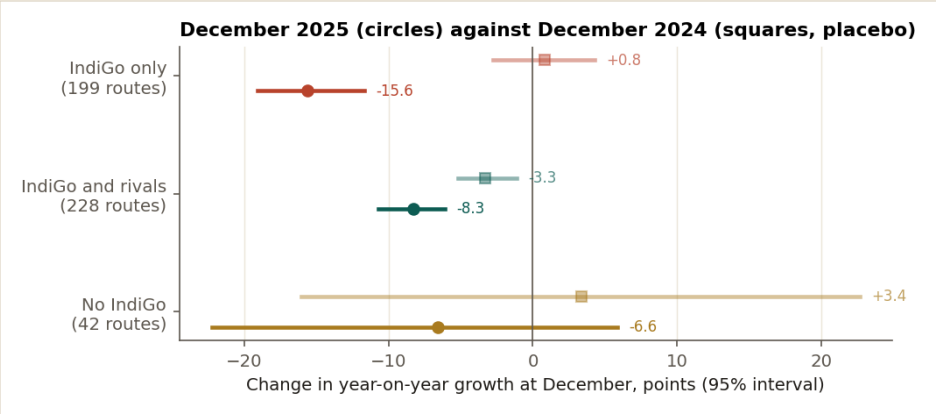


FIGURE 0.4 Change in year-on-year growth at December, by route group: 2025 (circles) and the 2024 placebo (squares), with 95% intervals from resampling routes.

ROUTE GROUP (ROUTES)	DECEMBER 2025	95% INTERVAL	PLACEBO 2024
IndiGo only (199)	−15.6	−19.1 to −11.7	+0.8
IndiGo and rivals (228)	−8.3	−10.7 to −6.1	−3.3
No IndiGo (42)	−6.6	−22.3 to +5.9	+3.4
H1: IndiGo only minus No IndiGo	−9.1	−21.7 to +7.1	−2.6
H2: IndiGo only minus IndiGo and rivals	−7.4	−11.4 to −2.7	+4.1

Points of year-on-year growth. Source: numbers.json, keys h_* and placebo_*.

Against the plan. H2 is supported: routes only IndiGo flew fell 7.4 points more than routes it shared, the interval excludes zero. The placebo year, which should show no gap, shows +4.1 points with an interval of −0.2 to +8.4: that includes zero, but only just. H1 is not supported by the pre-registered rule: the routes IndiGo did not fly are only 42, mostly small regional ones, and their interval is too wide. By the pre-registered rule the placebo passes on both: comfortably for H1, narrowly for H2.

How robust is H2? It is larger on busier routes (−10.3 points, interval −15.5 to −4.8) and points the same way in every check, but three checks cannot exclude zero: unweighted (−2.8), routes where both airports' articles agree (−5.1) and routes away from Delhi, Mumbai and Goa (−4.8). The gap is a fact about where passengers were, not about every small route. A fixed-effects regression with standard errors clustered by route gives the same point estimates.

source: tables/route_robustness.csv

Is the gap passengers switching airlines? Probably not. Taking IndiGo's November passengers, minus those on IndiGo-only routes, over all passengers on shared routes puts its share of shared routes at about 59.6%. If shared routes simply lost IndiGo's share of what IndiGo-only routes lost, they would have fallen 9.3 points; they fell 8.3. The difference leaves little room for switching, which matches the national finding. Within shared routes, those with one rival group fell 10.6 points and those with two or more 7.1. (Exploratory; see DEVIATIONS.md.)

PASSENGERS MISSING AGAINST EACH ROUTE'S OWN TREND, DECEMBER 2025

Expected December 2025 passengers are each route's December 2024

6 How long did it last?

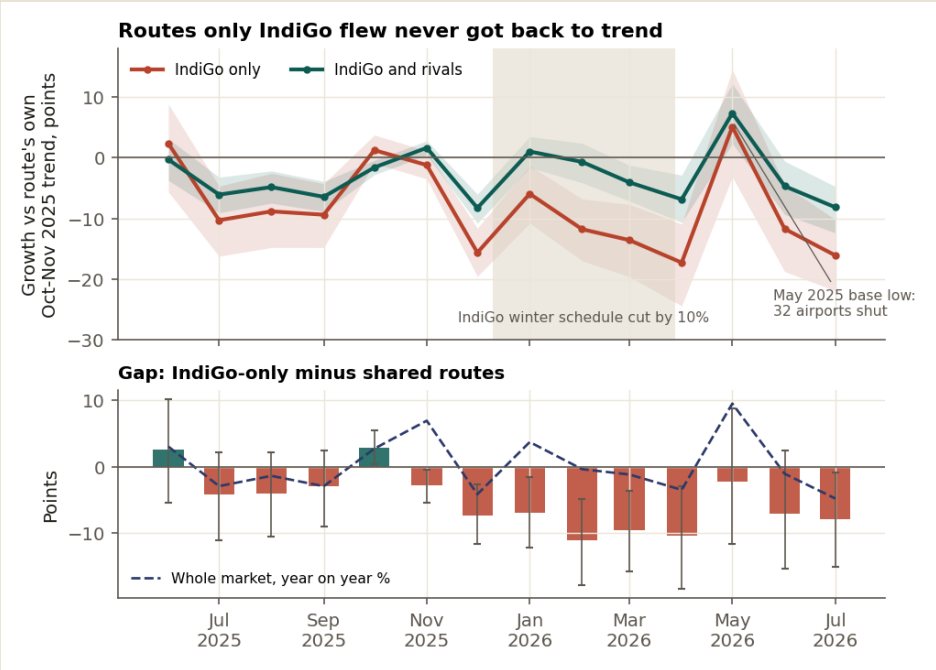


FIGURE 0.5 Top: each group's growth against its own October to November 2025 trend. Bottom: the gap between IndiGo-only and shared routes, with the whole market's year-on-year growth for context.

Shared routes were back on trend by January 2026. IndiGo-only routes were not. From February to July 2026 they ran on average 8.0 points behind shared routes, and in 4 of those six months the interval excludes zero. The 10% cut to IndiGo's winter schedule ran to the end of March; IndiGo's shortfall from December to March, by the method of Section 3, was about 4.21 million passengers (exploratory).

> THE MAY 2026 ``RECOVERY'' IS THE MAY 2025 BASE
>
> The pre-registered rule says IndiGo-only routes recovered in May 2026. But in May 2025, 32 airports in northern and western India were shut to civil flights for several days during the India-Pakistan conflict, so May 2025 traffic on their routes was low and May 2026 growth looks high. Without the 67 routes touching those airports, May 2026 is −4.0 points against trend for IndiGo-only routes and +1.5 for shared ones.

7 Who came out stronger?

IndiGo's share averaged 64.5% from January to November 2025, fell to 59.6% in December, was back to 63.6% in January and reached a record 67.4% in July 2026. Concentration rose with it, from an average index of 4,904 to 5,144 in July (anything above 2,500 is "highly concentrated" in competition practice). The airline that failed its passengers ended the period with more of them.

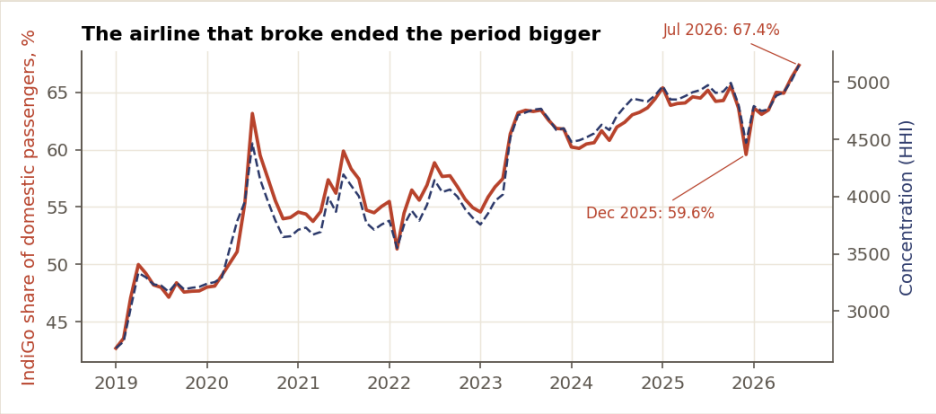


FIGURE 0.6 IndiGo's share of domestic passengers and the Herfindahl-Hirschman index of concentration, airline groups with the Tata carriers combined in every month.

8 Which cities had no backup?

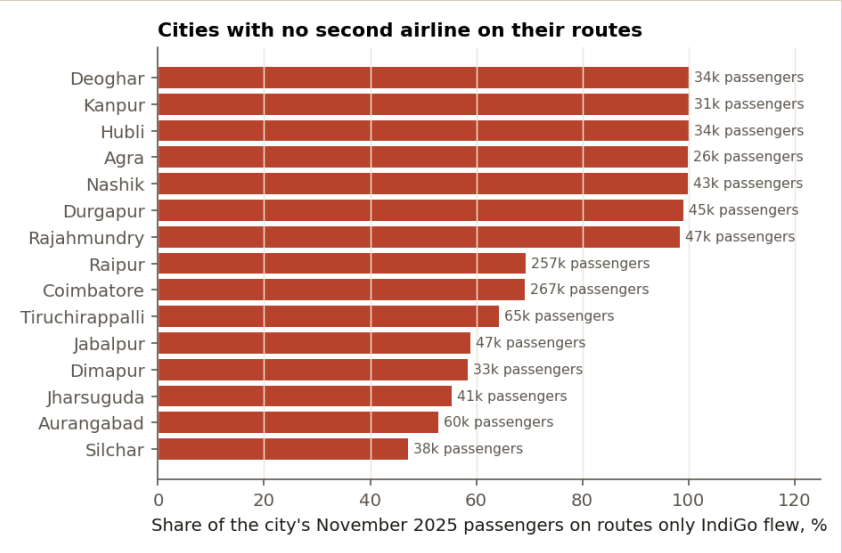


FIGURE 0.7 Share of each city's November 2025 passengers on routes where IndiGo was the only airline listed. Cities with at least 20,000 passengers.

In November 2025, 14.5% of domestic passengers flew routes where IndiGo was the only airline listed. For 14 of 71 cities with 20,000 or more passengers that was most of their traffic, and the more exposed a city, the more it fell in December (passenger-weighted correlation −0.27). Given the spot check, these shares are upper bounds city by city; the pattern, not any single city's figure, is the finding.

9 What this does not show

- Who flew each route is reconstructed from Wikipedia, checked against traffic but not against airline schedules; some shared routes are misclassified as IndiGo-only.
- Monthly data cannot isolate the worst week, and 4 to 6 December are miss-

ing from the daily archive.

- No fare data: the fare caps of December are context, not measured.
- 2026 had other shocks (the fuel-price spike from March, the schedule cut to March); comparing route groups with each other removes what hit the whole market, not what hit them differently.
- Not peer reviewed.

10

Method in one paragraph

Pinned downloads with SHA-256 checksums; a city-name table; a Wikipedia parser with tested rules for “begins”, “ends”, seasonal and charter entries, reading fixed revisions; a dbt warehouse on DuckDB with 18 tests; passenger-weighted group means with 2,000-draw route bootstrap intervals (seed 20251201); a placebo year; six pre-specified robustness checks; every number in this report generated from `results/numbers.json`. The plan was committed before any December result by route group was computed; `DEVIATIONS.md` lists five bug fixes and every addition. `make all` regenerates everything.